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1. What is LLM?
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**** LLM = Large Language Model**

**** An LLM is an AI model trained on a huge amount of data to understand and generate human-like language.**

Ex:

- 1) Open AI GPT
- 2) Google Gemini
- 3) META LLama
- 4) DeepSeek R1
- 5) Anthropic Claude

**** An LLM is a machine learning model that can read text, understand context, and generate meaningful responses.**

**** It can perform tasks like:**

- 1) Answer questions
- 2) Write Emails
- 3) Documents Summarization
- 4) Translate Languages
- 5) Generate Code
- 6) Explain Code
- 7) Generate Images
- 8) Generate Audios and Videos
- 9) Generate PPT

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Normal Python Program
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```
name = "Ashok"
print(name)
```

Output : Ashok  
```

Note : This program gives output only based on fixed instructions.

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LLM-Based Program
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User Prompt : Explain Python list with examples

LLM Response : A list in Python is used to store multiple values in a single variable...

```
list = [10, 20, 30, 40]
```

Note : Here, the LLM understands the question and generates a meaningful answer based on the context.

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How LLM Works Internally
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Step-1 : User input / User Prompt

Step-2 : Tokenization

Step-3 : LLM Execution

Step-4 : Predict Next token

Step-5 : Generate Response

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What is Tokenization?
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==> LLMs do not directly understand words like humans.

==> They break text into smaller pieces called tokens.

Input: I love Python

Tokens: ["I", "love", "Python"]

Note: Sometimes one word can be split into multiple tokens.

Input: Generative

Tokens: ["Gener", "ative"]

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What Does an LLM Actually Do?
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** An LLM predicts the next token based on previous tokens.

Input: Python is a programming

Prediction: language

The model keeps predicting tokens one by one until the answer is complete.

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Why Are LLMs Called "Large"?
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** They are called large because they are trained using:

- 1) Huge Data sets
- 2) Billions or Trillions of words
- 3) Large Number of parameters
- 4) Powerful Systems with GPU
- 5) Advanced Deep Learning Architecture

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LLM vs Traditional Software

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- Traditional software works based on Rules or instructions
- LLMs learns from data
- Traditional software provides fixed output
- LLM generates dyanmic ouput based on context
- Traditional softwares needs extra logic for processing
- LLM understand Natural language
- Traditional software example is : Calculator
- LLM example is : ChatGPT

Note: LLMs do not "think" exactly like humans.

**** They generate responses based on patterns learned from training data and the context provided in the prompt.

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Paid LLM vs Free LLM

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** A paid LLM is usually accessed through an API or subscription.

Example :

- 1) OPEN AI GPT Models
- 2) Anthropic Claude Models
- 3) Google Gemini
- 4) Cohere enterprise model

** we will usally pay based on

- 1) Number of tokens we are using
- 2) Monthly Subscriptions
- 3) API usage

4) Enterprise plan

** Paid LLM Advantages

- 1) Easy To use
- 2) Better Performance
- 3) No Infrastructure Mgmt for us
- 4) Enterprise Features

** Paid LLM Dis-Advantages

- 1) Cost increases with usage
- 2) Internet speed required
- 3) Vendor dependency
- 4) Data Privacy
- 5) Rate Limits may apply (RPM)

** Free LLM means

- 1) Free web access with limits
- 2) Free API access with limited usage
- 3) We can download them and run in our local machine

Free LLM Examples

- 1) Free ChatGpt access
- 2) Free Gemini Access
- 3) Llama models
- 4) Mistral models
- 6) DeepSeek Models

Free / Open-source LLM Advantages

- 1) Low cost
- 2) Better Control
- 3) Privacy
- 4) Customize

Free / Open-source LLM Disadvantages

- 1) Limited Results
- 2) Slow performance
- 3) Maintenance

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GEN AI Project Development using OPEN AI GPT Model

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Step-0 : Create Account in OPEN AI Platform + Make Payment + GENERETE API Key

URL : <https://platform.openai.com/>

Step-1 : Create project folder using IDE

Ex : 01-openai-gpt-chat-bot

Step-2 : Create Python Virtual Environment and activate it using below commands

\$ cd 01-openai-gpt-chat-bot

\$ python -m venv venv

\$ venv\Scripts\activate

Step-3 : Create "requirements.txt" and configure libraries and install them using pip

...

openai
python-dotenv

...

Step-4 : Create .env file and store openai key

Step-5 : Create "main.py" to consume GPT LLM

input : prompt
output : LLM Response

Step-6 : Create app.py with streamlit ui (like chatgpt ui)

Step-7 : Run the applicationa and test it.